

Param Shah

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EDUCATION

University of Waterloo

Honours Bachelor of Mathematics

Joint Honours Statistics and Computing

Waterloo, Canada

Sep 2022 – Dec 2027

- **Relevant Courses:** Object-Oriented Programming, Data Structures and Algorithms, Machine Learning, Database Systems, Portfolio Optimization Models, Derivatives, Stochastic Processes, and Linear Algebra.

EXPERIENCE

Trading and Research Intern

Capula Investment Management

Jun 2026 – Aug 2026

London, UK

- Researched systematic strategies around French OAT auctions on the Fixed Income RV desk; built an event study over 1,400+ auctions and 20 years of data measuring pre-auction concession in futures and across the auctioned line and its neighbours, quantifying overbidding, and valuing the embedded greenshoe (ONC) option conditional on auction tail, bid-to-cover and issuance size.
- Priced the ONC greenshoe as a one-day European call struck at the auction weighted average price, delta hedging in OAT futures to expiry to attribute P&L between gamma and hedging error against realized post-auction vol.
- Shipped the framework as a dashboard now used as standard tooling by the Euro FI RV desk's three researchers, running ahead of every upcoming auction and feeding pre-auction positioning.

Quantitative Developer Intern

Causeway Capital Management LLC

May 2025 – Aug 2025

Dallas, TX

- Productionized short-squeeze signal triangulating short interest, utilization and borrow data to flag crowded shorts, deployed as a daily screen in the PM portal; it flagged 70% of squeeze events, defined as 20%+ five-day moves.
- Built a data pipeline and orchestration platform on Airflow and FastAPI ingesting multi-source market data on a daily schedule, with internal dashboards giving real-time visibility into pipeline health and production runs, cutting deployment time 60%.
- Prototyped an LLM agent pipeline ingesting cross-asset market data to classify equity regimes via tool-augmented reasoning over price and macro feeds, delivering daily regime tags to downstream signal models.

President, Quantitative Research

UW Stocks Club

Jan 2025 – Jan 2026

Waterloo, Canada

- Led a 25-member team to launch an open-source Python toolkit for event-driven algo trading with modular broker adapters, real-time risk analytics, and plug-and-play strategy templates (Fama-French 5, Black-Litterman, SMA).
- Architected a Streamlit back-testing framework with tick-level rewind, walk-forward optimization, and portfolio dashboards, cutting validation cycles by 90% and becoming the club's go-to engine for rapid strategy testing.

Software Engineer Intern

Siemens Healthineers

Jan 2025 – Apr 2025

Ottawa, Canada

- Integrated advanced Android Debug Bridge (ADB) workflows, including real-time log capturing, crash diagnosis, and automated device provisioning, reducing mobile testing turnaround by 50%.

QA Automation Engineer Intern

Axonify Inc.

May 2024 – Aug 2024

Waterloo, Canada

- Streamlined automated testing workflows by deploying test environments with Docker and Kubernetes, reducing environment setup time by 35% and ensuring consistency across multiple test environments.

PROJECT

Football (Soccer) Match Data Pipeline & Betting-Edge Model

- Building a Python pipeline that scrapes post-match football data (xG, shot maps, lineups, odds) via Playwright and BeautifulSoup into PostgreSQL on a scheduled Airflow DAG, then models outcomes to estimate fair odds against bookmaker lines and size value bets via fractional-Kelly.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Javascript, SQL

Quant & Data Science: NumPy, pandas, SciPy, statsmodels, scikit-learn, PyTorch

AI/LLM: LangChain, LlamaIndex, RAG pipelines, LLM agents & tool-augmented reasoning

Data & Infrastructure: PostgreSQL, Redis, Airflow, Kafka, FastAPI, Docker, Kubernetes, AWS, Git, Web Scraping